ARCHITECTURE DECISION RECORDS (ADR - SHEET 1)

ADR-001: PLATFORM SELECTION — ROCKCHIP RV1103 OVER ESP32-S3

• **Status:** Accepted | **Date:** 2026-10-08

• **Context:** ESP32-S3 software-encodes MJPEG video, saturating the CPU and PSRAM bus and causing severe frame drop stutters when transmitting over Wi-Fi.

• **Resolution:** Adopt the **Luckfox Pico Mini B (Rockchip RV1103)** equipped with dedicated hardware ISP 3.2 and hardware H.264 VPU compression.

• **Technical Rationale:** Hardware video pipeline operates at zero CPU overhead. Produces smooth 30.0 fps Full HD H.264 video with clean RTSP streaming.

ADR-002: STORAGE TOPOLOGY — ZERO SD CARD EXECUTION (128MB SPI NAND)

• **Status:** Accepted | **Date:** 2026-10-08

• **Context:** High-endurance 32GB SD cards cost upwards of INR 800.00 and represent the primary failure point under continuous vehicular writes.

• **Resolution: Eliminate external SD card entirely.** Boot and execute firmware exclusively from onboard 128MB SPI NAND Flash.

• **Technical Rationale:** Reduces BOM to INR 3,264.00 (under INR 3,800 ceiling). Eliminates filesystem corruption on sudden ignition power loss. Live stream is recorded on demand directly inside the client smartphone application (e.g. VLC).

ADR-003: VIDEO PROTOCOL — HARDWARE H.264 RTSP VIA WI-FI SOFTAP

• **Status:** Accepted | **Date:** 2026-10-08

• **Context:** Video stream must transmit reliably to smartphones without requiring an external Wi-Fi router or cellular internet connectivity.

• **Resolution:** Host a local IEEE 802.11n Access Point (SoftAP) via an RTL8188EUS USB adapter and stream H.264 over RTSP/RTP (Port 554).

• **Technical Rationale:** Reduces network bandwidth to ~4 Mbps (vs 25+ Mbps for MJPEG). Compatible with standard mobile video players (VLC).

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT	SIZE A4	DOCUMENT NUMBER ENG-DASH-ADR-001	DATE 2026-10-08	SHEET 1 OF 2	
DOCUMENT TITLE ARCHITECTURE DECISION RECORDS	REV 1.4	DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824		
		ENGINEERING DISCIPLINE ARCHITECTURE & SYSTEMS ENGR	STATUS RELEASED		

ARCHITECTURE DECISION RECORDS (ADR - SHEET 2)

ADR-004: OPTICAL SENSOR SELECTION — SMARTSENS SC3336 3MP MIPI CSI-2

• **Status:** Accepted | **Date:** 2026-10-08

• **Context:** Selecting an image sensor compatible with the RV1103 hardware ISP without requiring driver modifications.

• **Resolution:** Adopt the **SmartSens SC3336 3MP MIPI CSI-2 Module** with 98.3 degree diagonal field-of-view lens.

• **Technical Rationale:** Fully supported by official Luckfox SDK with pre-tuned Image Quality (IQ) calibration binaries for automotive scenes.

ADR-005: POWER MANAGEMENT & INPUT PROTECTION CIRCUIT TOPOLOGY

• **Status:** Accepted | **Date:** 2026-10-08

• **Context:** Vehicle 12V rails exhibit severe electrical transients, load dumps up to 40V, and inductive spikes. Furthermore, the IP5306 power management IC automatically sleeps if load drops below 50mA.

• **Resolution:** Implement a robust multi-stage protection and conditioning network:

PROTECTION STAGE	COMPONENT SELECTION	CIRCUIT FUNCTION & VERIFICATION
Overvoltage Surge Suppression	SMBJ5.0A TVS Diode (D1)	Shunts vehicle alternator load dump spikes and reverse polarity transients safely to ground.
Bulk Reservoir Damping	470uF / 25V Electrolytic (C5, C6)	Absorbs vehicle supply line ripple and buffers Wi-Fi packet transmit current spikes.
High-Frequency Decoupling	100nF Ceramic (C1, C2, C3, C4)	Suppresses high-frequency switching hash across buck regulator, PMIC, SBC, and Wi-Fi rails.
Anti-Sleep Keep-Alive Circuit	100 Ohm Resistor (R1)	Enforces a static ~50mA pull on IP5306 output to permanently disable the 32-second auto-shutdown timer.

ORGANIZATION / PROJECT DASHCAM PROTOTYPE PROJECT		SIZE A4	DOCUMENT NUMBER ENG-DASH-ADR-001	DATE 2026-10-08	SHEET 2 OF 2
			DESIGN ENGINEER Souvik H	ENGINEER REGISTRATION NO. IN-ENG-2022-22824	
DOCUMENT TITLE ARCHITECTURE DECISION RECORDS		REV 1.4	ENGINEERING DISCIPLINE ARCHITECTURE & SYSTEMS ENGR	STATUS RELEASED	